THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET A

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The next number in the square number sequence $1, 4, 9, 16, 25, \ldots$ is: 1

(a) 38 **(b)** 36 **(c)** 40 **(d)** 37

2. The next number in the odd number sequence $1, 3, 5, 7, 9, \ldots$ is: 1

(a) 11 **(b)** 10 **(c)** 13 **(d)** 12

3. The 8th square number is: 1

(a) 72 **(b)** 64 **(c)** 56 **(d)** 81

4. Which of these is NOT a square number? 1

(a) 50 **(b)** 64 **(c)** 81 **(d)** 49

5. The term after 21 and 34 in the Virahānka sequence is: 1

(a) 56 **(b)** 65 **(c)** 55 **(d)** 45

6. The Koch snowflake is built by repeatedly: 1

(a) adding a ‘speed bump’ to each line segment
(b) joining midpoints
(c) removing corners
(d) doubling every angle

7. Which sequence contains only the number 1 repeated? 1

(a) even numbers **(b)** cubes
(c) counting numbers **(d)** the ‘all 1’s’ sequence

8. A rule that tells how to get each term of a sequence is called a: 1

(a) fraction **(b)** pattern rule **(c)** perimeter **(d)** angle

9. The 9th triangular number is: 1

(a) 46 **(b)** 45 **(c)** 54 **(d)** 43

10. The sum of the first 12 odd numbers is: 1

(a) 24 **(b)** 132 **(c)** 156 **(d)** 144

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the first five square numbers. 2

12. Write the first five triangular numbers. 2

13. Why are the numbers 1, 4, 9, 16, … called square numbers? 2

14. Find the sum $1 + 3 + 5 + 7 + 9$ without adding one by one. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain with a picture why the sum of the first n odd numbers is n^2. 3

16. Explain why 1, 3, 6, 10, … are called triangular numbers. 3

17. Describe the Virahānka (Fibonacci) sequence and write its first 10 terms. 3

OR

Find the sum $1 + 2 + 3 + \cdots + 10$ using a pattern. Explain. 3

18. Write the rule and the next three terms of 4, 7, 10, 13, … 3

OR

Draw the next shape in a sequence of squares with 1, 4, 9 dots and say how many dots it has. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Write the first eight triangular numbers. 5
(ii) Show that two consecutive triangular numbers give a square.
(iii) Draw the 4th triangular arrangement.

OR

(i) Define square numbers and write the first ten. 5
(ii) Show visually that 1+3+5+7 = 16.
(iii) Find the sum of the first 12 odd numbers.

20. (i) What is the Virahānka sequence? 5
(ii) Write its first twelve terms.
(iii) Where do you see this in nature?

OR

(i) Write the sequences of cubes and powers of 2 up to six terms. 5
(ii) Which grows faster?
(iii) Give the 10th term of each.

END OF QUESTION PAPER — Total: 40 Marks

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